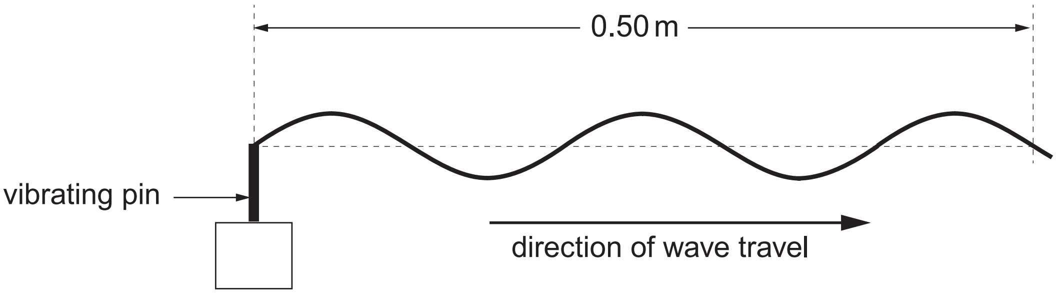
Examiner only

4. *(a)* A **progressive** wave generated by a vibrating pin is travelling on a long string. The periodic time is 0.040 s. The diagram shows a part of the string at time t = 0.



(i) Calculate the distance that the wave travels in a time of 0.34 s. [3]

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(ii) Carefully sketch **on the diagram above** the same part of the string at time t = 0.34 s. (The periodic time is 0.040 s.) [1]

(b) If the string above is clamped rigidly a little way to the right of the part shown, a **stationary** wave may be observed.

(i) State what is meant by a **node** in a stationary wave, and state how far apart the nodes will be in *this* stationary wave. [2]

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(ii) State how, if at all, the *phase* of the oscillations of the particles of the string varies with distance along the string, for a stationary wave. [2]

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Examiner only

(iii) Explain, in terms of *interference*, how the stationary wave is produced for this stationary wave on the string. [2]

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(iv) Stationary waves may also be observed on the string when the wave source is set to higher frequencies. State **one** way, apart from having a different frequency, in which the stationary waves will be different from the original stationary waves. [1]

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